MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

© 1/ TYPES 6SA7, 6SA7WGT, 12SA7, 12SA7GT, AND 12SY7
a b c d e

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentagrid converter

Outline	a, c, e	- - -	8-1 (EIA)
	b, d	- - -	9-11 or 9-41 (EIA)
Base	a, c, e	- - -	B8-21
	b, d	- - -	B8-6 or B8-58
Envelope	a, c, e	- - -	MT8
	b, d	- - -	T9
Cathode	All	- - -	Coated unipotential

Base connections:

Pin No.			1	2	3	4	5	6	7	8
Element	a, c, e	- - -	g5, sh	h	a	g2, g4	g1	k	h	g3
Element	b, d	- - -	nc	h	a	g2, g4	g1	k, g5	h	g3

ABSOLUTE-MAXIMUM RATINGS:

Parameter: / Unit:		Ef V	Eb Vdc	Ec2+4 Vdc	Ec1 Vdc	Ec3 Vdc	Pp W	Pg2+4 W	Ic1 mAdc	Ik mAdc	Ehk v	Alt ft
Maximum:	a, b	6.9	330	110	---	---	1.1	1.1	0.5	15.5	100	Note 1
Minimum:	a, b	5.7	---	---	---	---	---	---	---	---	---	---
Maximum:	c, d, e	13.9	330	110	---	---	1.1	1.1	0.5	15.5	100	Note 1
Minimum:	c, d, e	11.3	---	---	---	---	---	---	---	---	---	---
TEST CONDITION (1):	a, b	6.3	250	100	0	-2	---	---	---	---	---	---
	c, d, e	12.6	250	100	0	-2	---	---	---	---	---	---
TEST CONDITION (2):	e	12.6	28	28	0	-1	---	---	---	---	---	---

GENERAL:

© Qualification - Not required

1/ To identify those tests that are applicable to a given type or to several types; tube types are designated by letters.

© denotes changes

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1326	Conversion trans-conductance (1)	All	Test condition (1) (see note 2 or 3)	1.0	II	Sc	300	600	..mhos
1266	Total grid current	a, b, c, d	Test condition (1) (see note 2 or 3 and note 4)	1.0	II	Ic3	0	-2.0	..Adc
1266	Total grid current	e	Test condition (2); Rg3 = 0.1 Meg (see note 2 or 3 and note. 4)	1.0	II	Ic3	0	-2.0	..Adc
1266	Grid emission	All	Preheat with Ehk = 100 Vdc for 5 minutes; Ec1 Ec3 -3.4 Vdc (see note 5)	---	---	Ic1	---	-5.0	..Adc
1237	Oscillator grid current (1)	All	Test condition (1) (see note 3)	1.0	II	---	---	---	---
1231	Emission	All	Eb Ec1 Ec2 Ec3 Ec4 30 Vdc (see note 4)	1.0	II	Is	70	---	mAdc
1256	Electrode current (anode)	All	Test condition (1) (see note 2 or 3)	1.0	II	Ib	2.5	4.5	mAdc
1256	Electrode current (screen)	All	Test condition (1) (see note 2 or 3)	1.0	II	Ic2,4	6.5	11.5	mAdc
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	All	Rp = 10,000 ohms; Ec1 = Ec3 = -3.5 Vdc	---	---	Ep	---	300	mVac
1301	Heater current	a, b		---	---	If	275	325	mA
1301	Heater current	c, d, e		---	---	If	138	162	mA
1336	Heater-cathode leakage	All		---	---	Ihk	---	20	µAdc
1326	Conversion trans-conductance (2)	e	Test condition (2); Ic1 = 120 µAdc (see note 2 or 3)	---	---	Sc	200	400	µmhos
1326	Conversion trans-conductance (3)	All	Test condition (1); Ec3 = -35 Vdc (see note 2 or 3)	---	---	Sc	0.5	25	µmhos
1306	Transconductance	All	Test condition (1); Eb = 100 Vdc; Ec3 = 0 (see note 6)	---	---	Sg1g2	3,500	5,900	µmhos
1237	Oscillator grid current (2)	e	Test condition (2); Rg1 = 50,000 ohms (see note 7)	---	---	Ic1	100	200	µAdc

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
1331	Direct-interelectrode capacitance	a, c, e	No shield	---	---	Cga	---	0.25	pF
		b, d	Shield No. 308			Cga	---	0.5	pF
		All	Shield No. 308 for T9 bulbs			Cg3-all	7.6	11.4	pF
		All	Shield No. 308 for T9 bulbs			Ca-all	7.0	12.0	pF
1041	Shock	b	450 G (see note 8)	15	S4	---	---	---	---
---	Post-shock test end points:								
1336	Heater-cathode leakage	b		---	---	Ihk	---	30	μAdc
1306	Transconductance	b		---	---	Sglg2	3,000	6,000	μmhos
1105	Permanence of marking	All		---	---	---	---	---	---
	Quality conformance inspection, part 3								
1501	Intermittent life	All	Test condition (1), Group A, Ehk 100 V (see note 9)	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours):								
1326	Conversion transconductance (1)	All		---	---	Sc	225	---	μmhos
1237	Oscillator grid current (1)	All		---	---	Icl	See note 10	---	---
© 1031	Vibration fatigue (see note 11)	b		---	---	---	---	---	---
---	Vibration-fatigue test end points:								
1336	Heater-cathode leakage	b		---	---	Ihk	---	30	μAdc
1306	Transconductance	b		---	---	Sglg2	3.000	6.000	μmhos
© 1031	Variable-frequency vibration (see note 11)	b	Rp 2,000 ohms	---	---	Ep	---	500	mVac

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude maximum peak voltage in the basic document.
2. Insert 20,000 ohms and a suitable bypass capacitor in series with grid No. 1. Apply sufficient ac signal to produce 0.5 mAdc average grid current.
3. In place of circuit of note 2 the tube may be operated in converter oscillator test set, with tank circuit impedance adjusted to 3,100 ohms, with Rgl 50,000 ohms, Icl shall be between 150 and 260 μAdc; or with Rgl = 20,000 ohms, Icl shall be between 380 and 640 μAdc. Ec3 0.
4. This test to be performed at the conclusion of the holding period.
5. One tube from every other day's production shall be subjected to this test. Any failure shall cause this test to be performed at AQL of 6.5, inspection level of S3.
6. Tie screen and anode together at socket.
7. Tank circuit impedance adjusted to 9.000 ohms.
8. Rated heater voltage shall be applied.

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NOTES: -Continued

9. Ec1 = -16 Vdc; Ec3 = 0; apply 16.5 Vac to grid 1. Ec1 may be obtained by self rectification.

10. See note 3. With Rg1 = 50,000 ohms; Ic1 (min) = 135 μAdc or with Rg1 = 20,000 ohms; Ic1 (min) = 330 μAdc.

© 11. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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Army - EL
Navy - EC
Air Force - 85

Review activities:
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Air Force - 99
DSA - ES

User activities:
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Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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